

Lab 35 - MCP Server Requirements

MCP Server - Requirements Specification

This document lists the requirements for building an MCP (Model Context Protocol) server. Implementation of the server, tools, and data files is out of scope for this document; the items below are the lab requirements only.

1. Runtime & Platform

Build the MCP server using JavaScript on Node.js.

Requirements:

- Use Node.js as the runtime.
- Implement the server in JavaScript (Node.js).
- Project should be runnable with npm scripts (e.g. start / dev).

2. Transport - HTTP Express Streamable

Expose the MCP server over HTTP using Express with streamable communication (Streamable HTTP transport), not stdio-only.

Requirements:

- Use Express as the HTTP framework.
- Use MCP Streamable HTTP communication (streamable HTTP transport).
- Clients must be able to connect to the MCP server over HTTP.

3. Tool - Access Users Dummy Data

Provide an MCP tool that returns or looks up users dummy data.

Requirements:

- Register an MCP tool dedicated to user dummy data access.
- The tool must read from a JSON dummy dataset for users (see Section 5).
- Support typical lookups (e.g. list users, get user by id).

4. Tool - Access Users Bank Account Dummy Data

Provide an MCP tool that returns or looks up users bank account dummy data.

Requirements:

- Register an MCP tool dedicated to bank account dummy data access.
- The tool must read from a JSON dummy dataset for bank accounts (see Section 5).
- Support typical lookups (e.g. list accounts, get accounts by user id).

5. JSON Dummy Data (for sections 3 and 4)

Create separate JSON dummy data files/datasets for each of the domains above:

5.1 Users dummy data (for Tool in Section 3)

- Provide a JSON file (or equivalent JSON dataset) of dummy users.
- Each user record should include fields such as: id, name, email, and optional profile fields.
- Include multiple sample users so tools can demonstrate list/get behavior.

5.2 Bank account dummy data (for Tool in Section 4)

- Provide a JSON file (or equivalent JSON dataset) of dummy bank accounts.
- Each account record should include fields such as: accountId, userId, accountType, balance, currency.
- Link accounts to users via userId so tools can query by user.
- Include multiple sample accounts across users.

Acceptance Checklist

- ☐ MCP server implemented in JavaScript / Node.js
- ☐ HTTP Express with streamable MCP communication
- ☐ Tool for users dummy data
- ☐ Tool for users bank account dummy data
- ☐ JSON dummy data for users
- ☐ JSON dummy data for bank accounts

Note: This PDF is the requirements source for lab_35_MCP. Do not treat this document as an implementation - only as the requirements list.